YEDİTEPE UNIVERSITY

Welcome to Mechanical Engineering

ME101/Freshman Orientation • 2026–2027
06/10/2026



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Department Chair • Mechanical Engineering

Faculty of Engineering & Natural Sciences • Yeditepe University

So… what is Mechanical Engineering?

Think of ME as the engineering of physical systems.

Solid systems

strength • motion • vibration

Fluid systems

pumps • flows • biofluids



Thermal systems

energy • heat • combustion

Control & automation

robots • sensors • actuators

Manufacturing + materials

processes • materials • products

Why choose Mechanical Engineering?

The degree keeps many doors open — while giving you a rigorous engineering core.

Broad engineering scope

ME connects mechanics, materials, thermal-fluid sciences, dynamics, control and manufacturing.

Creative problem solving

You turn physical principles into useful machines, products, processes and technologies.

Career mobility

Design, R&D, robotics, defense, aerospace, automotive, energy, manufacturing, HVAC and management.

Interdisciplinary by nature

Modern ME works with electronics, computer science, medicine, materials and data/AI.

A car is a Mechanical Engineering system

One product can bring together almost everything you will learn.



Powertrain	thermodynamics • dynamics
Chassis & structures	strength • vibrations
Brakes & suspension	dynamics • materials
Controls & electronics	sensors • control
Manufacturing	processes • quality
Aerodynamics	fluid mechanics

What will you actually learn?

The first years build the language; the later years use it to design and solve systems.

Mathematics

calculus • differential equations • linear algebra

Physics

mechanics • energy • physical intuition

Modeling

equations • assumptions • simulation

Design

CAD • machine elements • engineering decisions

Computing

MATLAB • Python • numerical methods • data

Experiment

measurement • uncertainty • validation

Communication

technical reports • presentations • teamwork

Engineering judgment

ethics • safety • cost • sustainability

Your ME roadmap

Courses are sequenced: each year gives you tools for the next.

YEAR 1

FOUNDATIONS

Calculus • Physics • Chemistry • Intro ME • Solid Mechanics

Build mathematical and physical intuition.

YEAR 2

CORE ME

Statics • Dynamics • Strength • Materials • Thermodynamics

Learn to model and analyze engineering systems.

YEAR 3

ANALYSIS & DESIGN

Fluid Mechanics • Heat Transfer • Machine Elements • Dynamics & Control • Manufacturing • Numerical Methods

Move from analysis toward engineering design.

YEAR 4

INTEGRATION

Instrumentation • Vibrations • Thermal Design • Project • Electives • Internship

Combine everything in open-ended engineering problems.

Source: Mechanical Engineering Department Curriculum & Prerequisites (01.09.2024)

Prerequisites Tree: Your Courses Are Connected

A prerequisite is more than a registration requirement — it is part of the sequence of your engineering education.

Examples of Course Chains

Mathematics
MATH 131 → MATH 132 → MATH 241

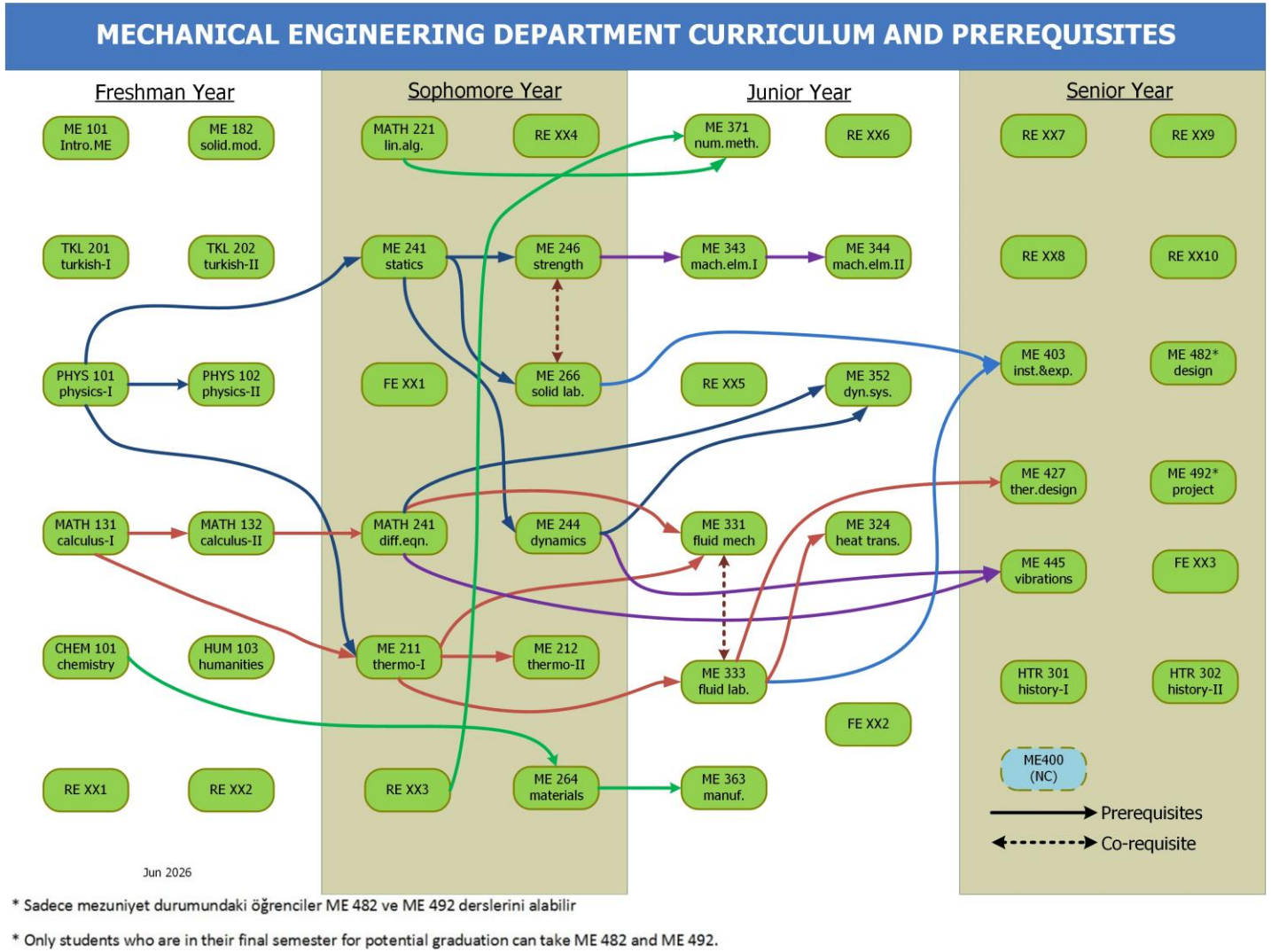
Thermodynamics
ME 211 → ME 212

Machine Design
ME 241 → ME 246 → ME 343 → ME 344

Dynamics & Control
MATH 241 / ME 244 → ME 352

Freshman Takeaway

- Know your prerequisites.
- Pass key courses on time.
- Plan your course sequence before registration.



* Sadece mezuniyet durumundaki öğrenciler ME 482 ve ME 492 derslerini alabilir

* Only students who are in their final semester for potential graduation can take ME 482 and ME 492.

Freshman year: build your foundation

Your goal is not to specialize yet. Your goal is to become comfortable thinking like an engineer.

ME 101	Introduction to Mechanical Engineering
MATH 131 / 132	Calculus I / II
CHEM 101	Chemistry
TKL 201 / 202	Turkish I / II
ME 182	Solid Mechanics
PHYS 101 / 102	Physics I / II
HUM 103	Humanities
RE XX1 / XX2	Required/elective curriculum components Academic English/Speaking (AFE 131/131, AFEA 111/112)

Freshman advice: understand the “why” behind equations — not just the steps to get an answer.

Source: Mechanical Engineering Department Curriculum & Prerequisites (01.09.2024)

Where can a Mechanical Engineer work?

The department's official description spans traditional and emerging sectors.

Design & Manufacturing

machines • products • production • quality

Energy & HVAC

thermal systems • power • heating/cooling

Automotive & Mobility

vehicles • dynamics • propulsion

Robotics & Automation

mechanisms • sensors • control

Defense & Aerospace

air/space vehicles • propulsion • structures

Biomedical & Biomimetics

devices • biomechanics • biofluids

Marine & Environmental

fluid systems • autonomous vehicles

R&D / Project / Management

research • product development • leadership

Research is where classroom knowledge becomes new technology

Undergraduate students can grow into research through projects, laboratories and faculty mentorship.

Robotics & Dynamics

parallel robots • legged locomotion • mechanisms • control

Biomechanics & Biomedical

gait • cardiovascular flows • FEA • medical devices

Thermal & Fluids

CFD • heat transfer • thermal systems • micro/nanofluidics

Aerospace & Propulsion

liquid rocket engines • satellite propulsion • aeroelasticity

Smart Materials & Energy

SMA • composites • vibration energy harvesting

Design & Manufacturing

CAD/FEA • machine elements • manufacturing processes

Faculty & teaching team

Your professors are researchers, designers, analysts and mentors — not just lecturers.

•**Prof. Dr. Koray K. Şafak** — Department Chair
Mechatronics, robotics, system dynamics & control, biomechanics

•**Prof. Dr. Mehmet A. Akgün**
Solid mechanics, structural optimization, composite structures

•**Prof. Dr. Hojin Ahn (Erdem An)**
Fluid mechanics, thermofluids, computational/experimental fluid dynamics

•**Prof. Dr. Ali Bahadır Olcay**
Thermofluids, heat transfer, fluid mechanics, cardiovascular flows

•**Prof. Dr. Nezih Topaloğlu**
Mechatronics, sensors/actuators, vibration and energy harvesting, MEMS

•**Assoc. Prof. Dr. Ali Fethi Okyar**
Biomechanics, solid mechanics, computational mechanics, biomedical devices

•**Dr. Namık Çıplak**
Robotics, mechanisms, kinematics

•**Dr. Onur Cem Namlı**
Solid mechanics, composite materials, material design and behavior

•**Dr. Mohammadamin Abdollahzadeh**
Solid mechanics, composite materials

Dr. Öğr. Üyesi Ahmet Ağaoğlu
Optimization • Additive Manufacturing • Mechatronics

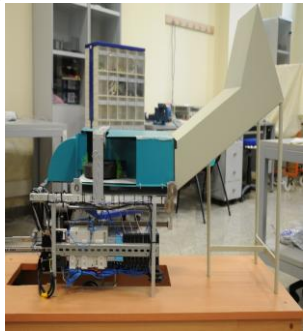
Lecturer Can Polat Çıgay
Control • Mechatronics • Robotics

Lecturer Aydin Akaltan
Design • Machine Elements

Faculty names/areas: updated department presentation (2026 version supplied by the department).

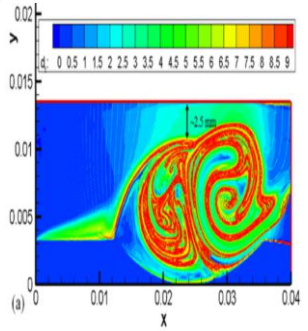
Labs: engineering is not only on the whiteboard

You will eventually measure, prototype, simulate and validate.



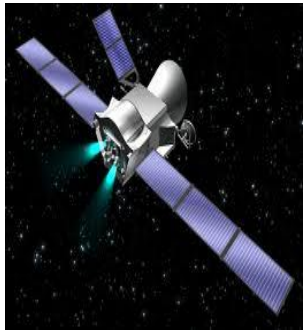
Robotics Research Lab

6-axis parallel micro-robots • legged locomotion • vision-guided automation



Solid Mechanics Lab

materials testing • fatigue • composites • shape-memory alloys



Fluid & Thermodynamics Labs

heat exchangers • micro-channel rigs • thermal imaging



Project & Research Lab

capstone projects • prototypes • student competitions

Facilities: updated department presentation; example images are from the supplied 2023 department deck.

Don't wait until senior year to become an engineer

The best undergraduate experiences usually happen outside the lecture notes.

Competitions

Teknofest, Shell Eco-Marathon, ASME-style challenges and student design competitions.

Capstone & projects

Build, test, fail, redesign — then explain why your solution works.

Industry exposure

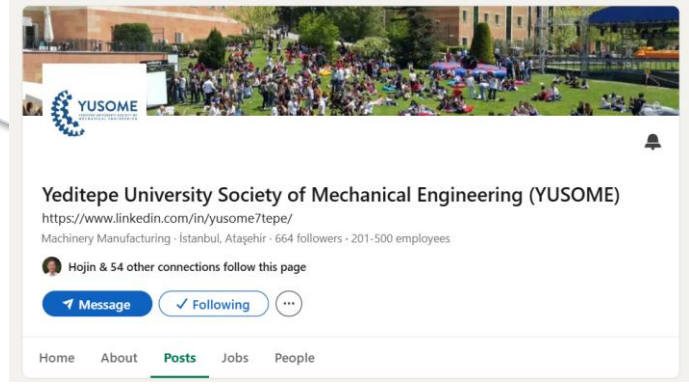
Department seminars, invited speakers and technical visits.

International opportunities

Erasmus / exchange and international academic experiences.

Student community

Join the departmental student body and professional societies.



Undergraduate research

Talk to faculty early if a research area catches your attention.

A program built around quality

Your degree is an internationally benchmarked engineering education.

MÜDEK

Reaccredited
30 Sep 2026 – 30 Sep 2029

EUR-ACE®

Internationally recognized
Bachelor label

100% ENGLISH

Undergraduate
program

Accreditation is not just a logo: it means the program is reviewed against defined engineering education standards and learning outcomes.

Four habits that will make your ME life easier

Start these habits now — before the courses become difficult.

1

Take active notes

Draw free-body diagrams. Write derivations. Ask what every term means.

2

Try before you search

Attempt the problem yourself before checking a solution, forum or AI tool.

3

Learn computational tools

Become comfortable with MATLAB and Python; add CAD and numerical methods as you progress.

4

Interact with people

Study with peers, use labs, attend office hours and talk to faculty about your interests.

Student guidance adapted from the supplied updated department presentation.

Use AI as a tool — not as a substitute for engineering

The department's AI guidance emphasizes learning, disclosure and academic integrity.

GOOD USES

Explain a concept after you have tried it
Debug MATLAB/Python/C/C++ code
Check an approach
Improve clarity and structure of writing

DISCLOSE

Include a brief declaration describing how AI was used.

Example: “AI was used for code debugging and language refinement. All analysis and results are my own.”

NOT OK

Generate prohibited full homework solutions
Fabricate lab reports or data
Invent plots/equations/results
Use AI during closed-book exams when prohibited

Rule of thumb: if AI did the thinking that your instructor expects you to demonstrate, you have gone too far.

Source: supplied updated department presentation • Fair Use of Artificial Intelligence (AI)

Your ME degree can lead to three broad journeys

Industry

Design • R&D • manufacturing • energy • automotive • defense • aerospace • robotics • HVAC

Research

Master's • PhD • laboratories • universities • advanced R&D

Leadership & innovation

Project management • engineering leadership • entrepreneurship • product development

You do not need to know your destination in your first semester. Your job now is to build a strong foundation and explore.

Start here: your first 13 weeks

A simple checklist for turning “I am an ME student” into “I am becoming an engineer.”

✓ Know your curriculum

Find prerequisites, grading rules and important academic dates.

✓ Build your math/physics base

Do not let small gaps accumulate.

✓ Learn one technical tool

Start MATLAB or Python and practice every week.

✓ Visit the department

Meet faculty, see the labs and learn what projects are happening.

✓ Join something

Student society, competition, project team, seminar or research activity.

✓ Ask early

If a course becomes difficult, ask for help before the problem compounds.

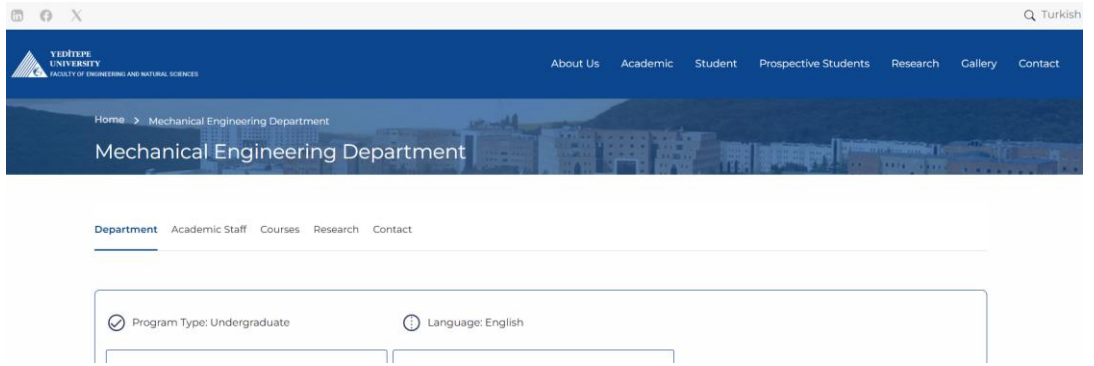
Welcome to the department — we are looking forward to seeing what you will build.

Sources of information

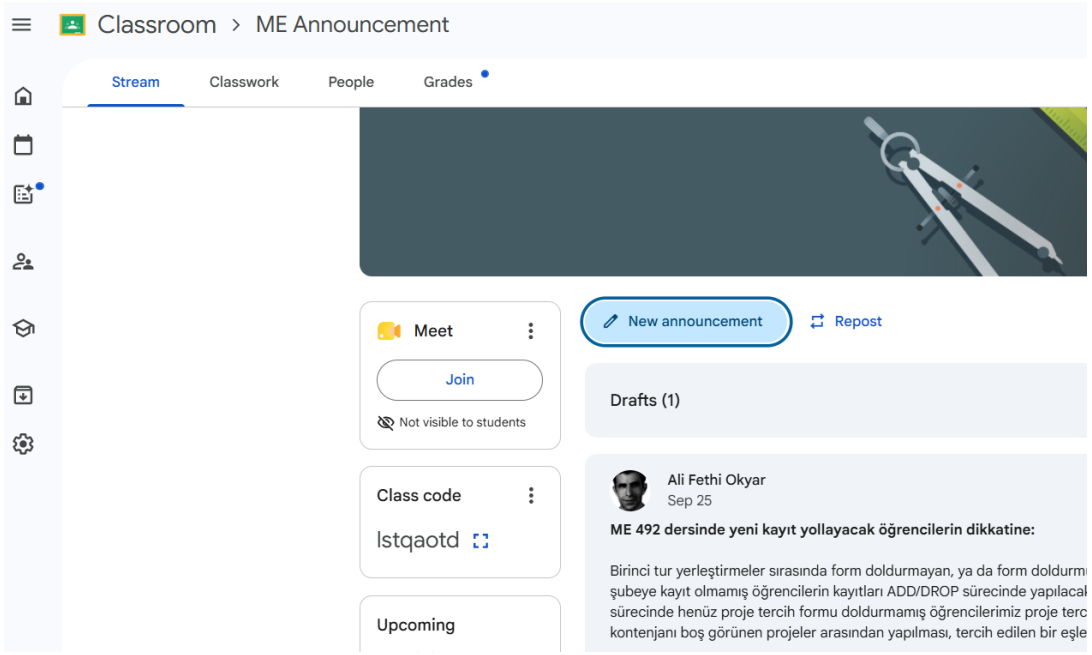
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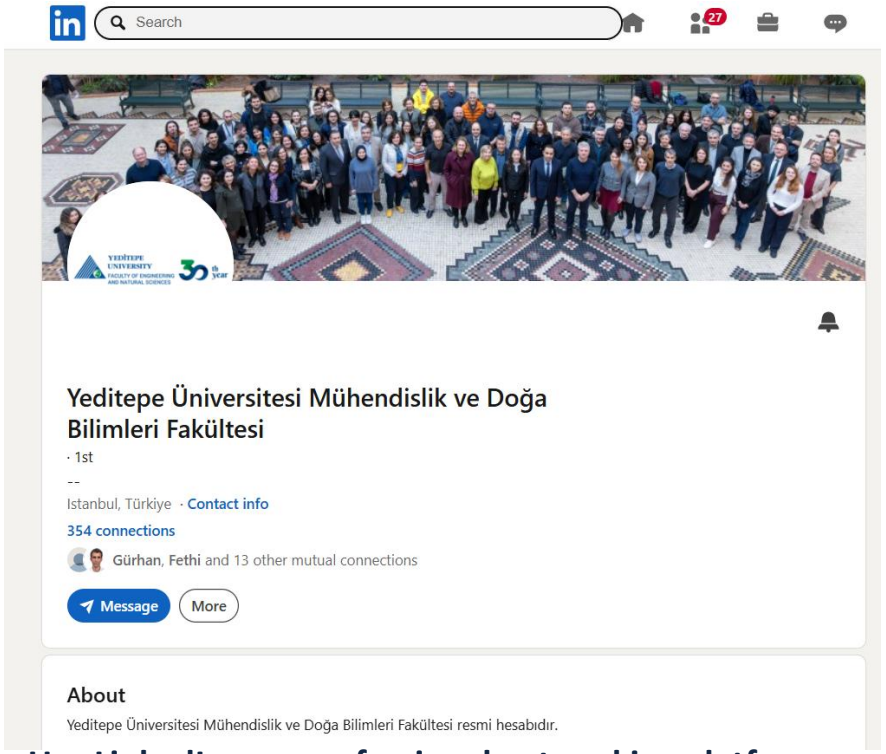
Check ME Dept. web page regularly:
https://mdbf.yeditepe.edu.tr/en/mechanical-engineering-department

- Program type, English instruction, department scope, curriculum areas, career fields, mission/vision.



Enroll in Classroom>ME Announcement





Use LinkedIn as a professional networking platform (even as a student)

Use Yeditepe mobile app

